

Variational Mixture of Stochastic Experts Auto-Encoder for Multi-Modal Recommendation

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Abstract—Multi-modal data presents a promising opportunity for improving multimedia recommendation models, but it also introduces task-irrelevant noise that can reduce model robustness. In this paper, we propose a robust multi-modal recommendation approach that accounts for different levels of task-irrelevant noise across modalities. We explicitly consider the uncertainty associated with each modality and perform stochastic sampling-based fusion according to the precision of different modalities, which serves as a measure of uncertainty. The influence of noisy modalities with high uncertainty is removed, filtering out task-irrelevant noise, and therefore a noise-robust multi-modal recommendation is achieved. Moreover, the stochastic sampling strategy intrinsically considers and simulates scenarios with absent modalities during multi-modal fusion. Consequently, it incorporates additional randomness into the training process, which enables the model to handle the problem of modality missing. Furthermore, the proposed fusion approach integrates the noise robustness of the Product-of-Experts (PoE) framework when modeling with Gaussian distributions, along with the flexibility of the Mixture-of-Experts (MoE) technique to represent diverse distributions of latent variables. This integration allows the proposed approach to achieve noise-robust modeling with non-Gaussian variables. Specifically, we derive a solvable evidence lower bound for the proposed variational mixture of stochastic experts (VMoSE) auto-encoder, where both Gaussian and Student-T distributions are used to model the latent variables. Constraints are added to match the similarities between the ID embeddings and the multi-modal joint embeddings by utilizing an Expectation maximization (EM)-style algorithm for better model optimization. Extensive experiments demonstrate the effectiveness of the proposed method in multi-modal fusion and the robustness to modality noise and modality missing.

Index Terms—Multi-modal recommendation, variational auto-encoder.

I. INTRODUCTION

MULTI-MODAL data fusion has become increasingly utilized in various applications, including multi-modal

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recommendation systems [1], [2] and multi-modal sentiment analysis [3], [4], to connect diverse data types and furnish supplementary data signals through shared information. Existing methods for multi-modal data fusion either learn a joint latent space for multiple modalities or model cross-association between different modalities while enforcing similarity constraints on corresponding latent vectors. Among the methods of learning joint variables, the MoE framework [5] is a commonly adopted methodology that averages the contribution of each modality equally. In contrast, PoE [6] weights them based on the uncertainty level of each modality, conferring greater robustness compared to MoE. MoPoE [7] trains multiple subsets of modalities to further augment the modeling capabilities of POE.

Despite the progress achieved by existing approaches in fusing multi-modal data, they inadvertently introduce task-irrelevant noise that can diminish model robustness. Specifically, although PoE demonstrates an ability to account for noise in individual modalities, it is analytically solvable only when modeling Gaussian latent variables. Models based on other latent distributions preclude closed-form solutions with PoE. Furthermore, the weighted summation of modalities according to their estimated inverse of variance is a consequence of Gaussian properties, rather than an explicit weighting aimed at enhancing robustness. Consequently, the PoE approach cannot model non-Gaussian latent variables and does not explicitly leverage the uncertainty levels of different modalities to better improve robustness.

In the context of multi-modal recommendation, typical methods [8], [9] often utilize multi-modal features extracted from pre-trained models (e.g., visual features from Resnet [10], textual features from BERT [11]) as input to uncover feature similarities for assisting recommendation. However, these pre-trained features often contain considerable task-irrelevant noise that cascades into the recommendation model, thereby reducing model robustness. Furthermore, user interactions are often driven by preferences for particular visual or textual content, rather than jointly by multi-modal item features. Therefore, identifying the key modalities that trigger user interactions and assigning proper weights, while filtering out the substantial task-irrelevant noise in modalities, is crucial for achieving robust multi-modal recommendations.

To address these challenges, we propose a novel multi-modal recommendation model termed Variational Mixture of Stochastic Experts (VMoSE). Specifically, we learn a Gaussian variable for each node, wherein the mean vector encapsulates semantic information, and the variance vector conveys the uncertainty or

noise level associated with the respective node. Then, we incorporate a sampling strategy that explicitly samples and fuses multi-modal features based on the estimated uncertainty level of the task. In this way, the proposed model also increases randomness during training, equivalent to randomly missing certain modalities while requiring the joint latent variable to reconstruct or predict, thereby improving the handling of modality missing. Additionally, by combining the advantages of both PoE and MoE, VMoSE can model non-Gaussian latent variables using a mixture of latent distributions. For instance, the Student-T distribution is a heavy-tailed distribution that has better robustness against outliers than Gaussian, making it more suitable for certain applications. Specifically, we design a corresponding probabilistic graph model for the recommendation system task and derive a solvable evidence lower bound (ELBO) to solve it using both Gaussian and Student-T based modeling. We also elucidate the physical interpretations of each term in the ELBO for the recommendation task.

The main contributions can be summarized as follows:

- We introduce a novel variational mixture model for multi-modal recommendations by adopting a stochastic approach to sample experts from various modalities. Specifically, we explicitly sample and fuse modalities based on their estimated uncertainty levels to the recommendation task, where the precision of each modality is used to measure the uncertainty. In this way, a noise-robust multi-modal recommendation could be achieved by removing the influence of highly uncertain modalities and filtering out task-irrelevant noise.
- The proposed stochastic sampling strategy of multi-modal features is leveraged to randomly omit certain modalities while forcing the joint latent variable to reconstruct or predict, which could be robust to modality missing. Moreover, by randomly sampling multiple modalities and mixing them, more flexible modeling with different distributions could be achieved. In this way, the proposed model can better handle the problem of non-analytical solutions of PoE under non-Gaussian distributions, enabling more flexible modeling.
- We design an appropriate probabilistic graphical model and utilize variational inference for solving it to achieve modality sampling and weighted fusion of multi-modal features. We further derive similarity constraints to enforce the resemblance between the multi-modal variable and ID variable, thereby leveraging the robust multi-modal variable to better assist the recommendation task. Experiments on three public datasets demonstrate the effectiveness of our proposed method over other baselines and the capability of better handling modality noise and missing.

The remainder of this article is organized as follows. Section II provides a review of related literature pertinent to this work. In Section III, we illustrate the proposed Variational Mixture of Stochastic Experts (VMoSE) model in detail, formally defining the methodology and underlying framework. Section IV presents the experimental settings and analyzes the experimental results. Finally, Section V briefly summarizes the article and suggests promising directions for future work.

II. RELATED WORK

A. Multi-Modal Fusion

Existing methods for multi-modal data fusion can be categorized into two primary methodologies: joint latent space learning [5], [6], [7] and cross-association modeling [12], [13]. The former aims to discover a shared representation that captures correlations among modalities. This is accomplished by training encoders to map inputs from different domains into a common embedding space. In contrast, cross-modal association modeling explicitly parameterizes inter-modality relationships via similarity constraints between embeddings from distinct modalities that correspond to the same entity. In this work, we focus on joint distribution-based methods.

Attention mechanisms have been widely adopted to learn dynamic and adaptive weights between modalities, and various recent works have developed attention models tailored for multi-modal data. For example, multi-modal transformer [4] presented a transformer model for multi-modal representation learning, which introduces a cross-modal self-attention mechanism to model intra- and inter-modal dynamics and dependencies for multi-modal sentiment analysis and emotion recognition. Sun et al. [14] proposed to adopt two neural co-attention mechanisms during image classification to help the model discover more complete object patterns for weakly supervised semantic segmentation (WSSS). The co-attention modules leverage cross-image semantic similarities and differences to complement intra-image information, improving object localization map quality and downstream segmentation performance. MAML [2] presented an attention module to transform multi-modal features into learned attention scores. These attention-weighted multi-modal representations are then fed into a metric learning module for making recommendations, which allows MAML to selectively emphasize the most relevant modalities and feature dimensions. These papers leverage self-attention and co-attention mechanisms to identify important interactions across modalities and integrate the modalities in a joint representation. However, attention offers a more general-purpose technique to dynamically reweight modalities based on content, rather than task relevance. Therefore, a specialized uncertainty-aware fusion approach using variational Bayesian principles should be explored.

Among variational-based methods, Suzuki et al. [15] proposed to concatenate encoded features of multiple modalities to learn the approximated joint posteriors. However, it cannot deal with modality-missing problems due to the concatenation. Shi et al. [5] approximated joint posteriors to follow a mixture of the single-modal variational posteriors through the average of individual variational posteriors. Wu and Goodman [6] proposed product-of-experts (PoE) to integrate distributions obtained from different modalities through a weighted average, where weights are reciprocal of variances of different modalities. Therefore, it could fuse useful information based on the reliability of the modalities. Sutter et al. [7] further extended PoE to incorporate different combinations of sub-modalities to better deal with modality missing. Lee and Pavlovic [16] introduced a disentangled multi-modal variational autoencoder (DMVAE) that utilizes a disentangled VAE strategy to separate the private

and shared latent spaces of multiple modalities. Mai et al. [17] proposed a multimodal information bottleneck (MIB) to learn the minimal sufficient representation for a given task by maximizing the mutual information between the representation and the target and simultaneously constraining the mutual information between the representation and the input data. Some other works are dedicated to utilizing a small amount of matching data for semi-supervised training [18], [19] and mitigating modality collapse [20].

However, few studies have focused on solving modality noise and uncertainty problems. Among them, Ma et al. [21] introduced a novel mixture of normal-inverse Gamma distributions (MoNIG) algorithm, which efficiently estimates the uncertainty in principle for the adaptive integration of different modalities and produces a trustworthy regression result. Compared to these methods, we first explicitly fuse different modalities according to their uncertainty level and leverage a stochastic sampling strategy for more randomness and robustness to the modality missing and noise.

B. Multi-Modal Recommendation

Self-supervised learning-based and graph-based methods have been widely utilized for multi-modal recommendations nowadays. Among them, MMGCN [22] showed a method that applies separate graph convolutional networks to model each modality. Recommendations are then derived by averaging the information from each modality-specific model to capture both intra- and inter-modal collaborative signals. Wang et al. [23] proposed to construct a user-user co-occurrence graph for mining the fusion preferences based on neighborhood collaborations, which could be utilized for modality missing through an attention-based fusion. Tao et al. [24] developed a self-supervised learning framework that applies data augmentation and contrastive techniques to uncover underlying relationships among modalities. It uses predictive pretext tasks to encourage the model to integrate and associate information across modalities in an unsupervised manner. Yi et al. [25] devised two data augmentation methods and proposed a novel negative sampling technique for graph-based modeling. Sun et al. [26] incorporated a multi-modal knowledge graph into recommender systems using graph attention modules. Peng et al. [27] proposed to leverage user-user graph, item-item graph, and user-item graph to conduct a multi-view graph convolution network for a more reliable multimedia recommendation. Zhang et al. [28] learned latent item-item semantic structures from multi-modal features through modality-aware graph learning and convolutions to enrich item representations for improved recommendations. Zhou et al. [8] proposed a self-supervised multi-modal recommendation model called BM3 that jointly optimizes three objectives to learn user and item representations by reconstructing the user-item graph and aligning multi-modal features, without needing negative samples or complex graph augmentations for contrastive view generation by simple dropout augmentation and stop-gradient. Wei et al. [9] proposed a multi-modal self-supervised learning method for recommendation that uses modality-aware interactive structure learning via adversarial

perturbations for data augmentation and cross-modal contrastive learning to jointly preserve inter-modal semantics and user preference diversity without relying on labels.

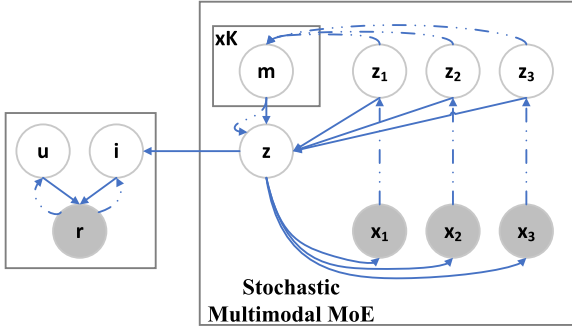
On the other hand, variational auto-encoder (VAE) [29] has also been applied in recommender systems due to their effectiveness at encoding high-dimensional data into latent probabilistic representations. The MultVAE model proposed by Liang et al. [30] incorporated a VAE with a multinomial likelihood model to fit user-item interaction histories. Specifically, a user's sequence of interacted items is provided as input to MultVAE, and item recommendations are made by reconstructing the multinomial probability distribution over items. MacridVAE [31] extended the VAE-CF framework by learning disentangled latent representations that capture users' preferences at both coarse-grained and fine-grained levels. For example, high-level latent attributes may represent a user's inclination toward broad item categories, while low-level attributes capture nuanced preferences such as clothing size. MacridVAE requires full interaction histories for some users to train the VAE effectively and generate recommendations for remaining users. More recently, Cho and Oh [32] proposed a stochastic collaborative filtering approach based on randomized expert VAEs. Their model samples a distinct expert VAE auto-encoder for each user-item interaction, such that a single user's clicks can be modeled by different experts' generative processes.

However, VAE has less been used in multi-modal recommendations. Among them, CVAE [33] presented a variational-based framework to incorporate different features into the modeling of the user-item collaborative information. Yi and Chen [1] proposed to utilize variational graph auto-encoders to model different modalities within the user-item bipartite graph, where different modality-specific latent variables are fused according to product-of-experts. Different from the above methods, our method further considers that the multi-modal features of items contain irrelevant information to the recommendation, and further optimizes the use of different noise levels in different modalities during fusion, so as to achieve more robust recommendations.

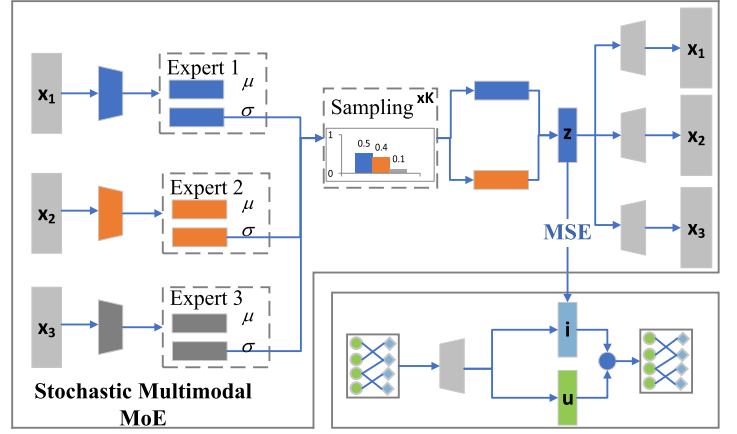
III. METHODS

A. Preliminaries

We consider a multi-modal recommendation problem with a dataset $\mathbf{R} = \{[r_{ui}]_{U \times I}, \mathbf{X}_{1:M}\}$ containing U users, I items and M modalities. r_{ui} denotes the rating of the i -th item by the u -th user, while $\mathbf{x}_{1:M}^{(i)}$ denotes the multi-modal features of the i -th item. The objective is to recommend uninteracted items to users that they are most likely to engage with. Specifically, we generate the predicted rating matrix $\hat{\mathbf{R}} \in \mathbb{R}^{U \times I}$ during testing and calculate the top- p highest predicted ratings of each user's uninteracted items to recommend. Notations used in this article are summarized in Table I. To distinguish between deterministic variables and random counterparts, we adopt the following notational convention: capital non-bold symbols (e.g., R) denote the random vector associated with the corresponding lowercase bold deterministic variable $\mathbf{r}$. Capital boldface symbols (e.g., $\mathbf{R}$) represent deterministic matrices.



(a) Probabilistic graphical model (PGM) of VMoSE.



(b) Framework of VMoSE.

Fig. 1. Overall structure of our proposed Variational Mixture of Stochastic Experts (VMoSE).

TABLE I
NOTATIONS AND DESCRIPTIONS

Notation	Description
$\mathcal{U}$	user set with U users
$\mathcal{I}$	item set with I items
d	dimension of latent variables
$\mathbf{u}$	latent user variable with the matrix of $\mathbf{U}$
$\mathbf{i}$	latent item variable with the matrix of $\mathbf{I}$
$\mathbf{z}_{1:M}$	latent item content variables with the matrix of $\mathbf{Z}_{1:M}$
$\mathbf{m}_{1:K}$	variables indicates which modalities to pick up
$\mathbf{z}$	latent item joint multi-modal variable
$\mathbf{r}$	rating vector with the matrix of $\mathbf{R}$
$\mathbf{x}_{1:M}$	content vector with the matrix of $\mathbf{X}_{1:M}$
U	latent user random variable
I	latent item random variable
$Z_{1:M}$	latent item content random variable
$M'_{1:K}$	K random variables indicates which modalities to pick up
Z	latent item joint multi-modal random variable
R	rating random variable
$X_{1:M}$	content random variable
θ_r, θ_x	generative parameters for rating, content variables
ϕ_{ui}, ϕ_z	inference parameters for rating, content variables
$p_{\theta_r}(R V)$	conditional likelihood of the ratings
$p_{\theta_x}(X Z)$	conditional likelihood of the content
$p(V)$	prior of the item random variable
$p(Z)$	prior of the item content random variable
$q_{\phi_i}(I R)$	variational posterior of item random variable
$q_{\phi_z}(Z X)$	variational posterior of item content random variable

B. Variational Mixture of Stochastic Experts Auto-Encoder

Given the effectiveness of probabilistic generative models for fusing multi-source information and the nonlinearity of VAE using neural networks to approximate the posterior and likelihood generation of encoders and decoders, and recognizing that VAE can well embed each node as a Gaussian variable (mean and variance) during training with the variance vector be well used to model uncertainty of the nodes caused by the sparsity of interactions and noise in the features, we employ variational inference for our generative model. In this section, we present the design of the probabilistic graphical model that defines the sampling and fusion process for the multi-modal item features. We also define an appropriate generative model to couple the multi-modal item

features with collaborative filtering information. The probabilistic graphic model and framework are depicted in Fig. 1.

1) *Generative Process*: VMoSE is a generative model where different modalities $\mathbf{x}_{1:M}$ are generated through modality-specific latent variables $\mathbf{z}_{1:M}$. Inspired by MAE [34], which greatly eliminates redundancy by sampling, and then constructs a task that cannot be easily solved, K variables $\mathbf{m}_{1:K}$ are sampled from a Categorical distribution to pick up K modalities, where a mixture of K modality-specific latent variables are fused to form the joint multi-modal latent variable $\mathbf{z}$. Specifically, multi-modal features $\mathbf{x}_{1:M}$ are generated from a set of latent variables according to the following generative process:

$$\begin{aligned}
 \mathbf{z}_m &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \\
 \mathbf{m}_k &\sim \text{Cat}(\pi_0), \\
 \mathbf{z}|\mathbf{z}_{1:M}, \mathbf{m}_{1:K} &\sim \frac{1}{K} \sum_{k=1}^K \prod_{m=1}^M [\mathbf{z}_m]^{\mathbf{m}_k^m}, \\
 \mathbf{x}_{1:M}|\mathbf{z} &\sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{z}; \theta_x), \text{diag}(\sigma^2(\mathbf{z}; \theta_x))), \quad (1)
 \end{aligned}$$

where $\mathbf{0}$ denotes the zero vector and $\mathbf{I}_d$ represents the d -dimensional identity matrix. Modality-specific latent variable $\mathbf{z}_m$ is sampled from standard Gaussian distribution. K denotes a predefined number of sampled modalities. A one-hot vector $\mathbf{m}_k$ is sampled from the Categorical distribution parameterized by π_0 to select one component from distinct modality-specific latent variables. We define $\pi_{0,m} = M^{-1}$ to render $\mathbf{m}_k$ uniformly distributed. $\mathbf{z}$ is a joint variable which is a mixture of K sampled modalities. The conditional probability distribution of each multi-modal feature $\mathbf{x}_m$ is Gaussian with mean $\mu_m(\mathbf{z}; \theta_x)$ and variance $\sigma_m^2(\mathbf{z}; \theta_x)$, where the parameters are generated by neural networks with weights θ_x .

To fully explore the collaborative information, we explicitly introduce the variables $\mathbf{i}^\dagger$ and $\mathbf{u}$ to embed the item collaborative information and user collaborative information, respectively. We model $\mathbf{i}^\dagger$ and $\mathbf{u}$ as draws from standard normal distributions, formally:

$$\mathbf{i}^\dagger \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \mathbf{u}^\dagger \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \quad (2)$$

Then, we represent the latent item variable $\mathbf{i}$ as the sum of the item collaborative variable $\mathbf{i}^\dagger$ and multi-modal joint latent variable $\mathbf{z}$ as follows:

$$\mathbf{i} = \mathbf{i}^\dagger + \mathbf{z}. \quad (3)$$

Given $\mathbf{z}$, the conditional distribution of $\mathbf{i}$ is normal with mean $\mathbf{z}$ and identity covariance matrix $\mathbf{I}_d$, or formally, $\mathbf{i}|\mathbf{z} \sim \mathcal{N}(\mathbf{z}, \mathbf{I}_d)$. This conditional dependence induces regularization between $\mathbf{i}$ and $\mathbf{z}$ under the Maximum a posteriori (MAP) objective, introducing mutual influence between the item's collaborative and multi-modal latent representations.

Finally, the observed user-item interaction sample $\mathbf{r}$ is modeled as being generated from the user and item embeddings, $\mathbf{u}$ and $\mathbf{i}$ respectively, through the following conditional Gaussian distribution:

$$\mathbf{r}|\mathbf{u}, \mathbf{i} \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{u}, \mathbf{i}; \theta_r), \text{diag}(\boldsymbol{\sigma}^2(\mathbf{u}, \mathbf{i}; \theta_r))), \quad (4)$$

where the mean $\boldsymbol{\mu}(\mathbf{u}, \mathbf{i}; \theta_r)$ and diagonal covariance matrix $\text{diag}(\boldsymbol{\sigma}^2(\mathbf{u}, \mathbf{i}; \theta_r))$ are parameterized by neural networks.

With the above generative process defined, we can write the complete joint distribution over all variables as:

$$\begin{aligned} p_\theta(Z_{1:M}, M'_{1:K}, Z, U, I, R, X_{1:M}) \\ = p(U)p(Z_{1:M})p(M'_{1:K})p(Z|M'_{1:K}, Z_{1:M}) \\ p(I|Z)p_{\theta_r}(R|U, I)p_{\theta_x}(X_{1:M}|Z), \end{aligned} \quad (5)$$

where θ_r and θ_x denote parameters with respect to the collaborative decoder and multi-modal content decoder, respectively. Capital non-bold variables such as R represent random vectors linked to their corresponding lowercase bold deterministic variables like $\mathbf{r}$, where $M'_{1:K}$ denotes the random variable of $\mathbf{m}_{1:K}$ to distinguish it from the total number of modalities represented by M . Here, each term refers to the corresponding variable's generative distribution, and this joint distribution encapsulates the complete generative process of all variables in the model. Specifically, $\{R, X_{1:M}\}$ are observed variables and $\{Z_{1:M}, M'_{1:K}, Z, U, I\}$ are hidden variables to be inferred. The joint distribution governing the item multi-modal features $p_\theta(Z_{1:M}, M'_{1:K}, Z, X_{1:M})$ is factorized as $p(Z_{1:M})p(M'_{1:K})p(Z|M'_{1:K}, Z_{1:M})p_{\theta_x}(X_{1:M}|Z)$. Here, $p(Z_{1:M})$ and $p(M'_{1:K})$ serve as prior distributions for modality-specific latent variables $Z_{1:M}$ and pick-up-modality indicator variables $M'_{1:K}$ (modeled as Categorical distributions). The conditional generative distribution $p(Z|M'_{1:K}, Z_{1:M})$ of the joint multi-modal variable Z is formulated as a mixture of distributions, incorporating different sampled modality-specific variables. The stochastic multi-modal content decoder $p_{\theta_x}(X_{1:M}|Z)$ is parameterized by neural networks, with θ_x to be trainable parameters. Concerning user-item interactions, the joint distribution $p_\theta(Z, U, I, R)$ is factorized as $p(U)p(I|Z)p_{\theta_r}(R|U, I)$. Here, $p(U)$ denotes the prior distribution for the user latent variable, and $p(I|Z)$ represents the conditional distribution of the item latent variable, which acts as a constraint coupling item multi-modal features and item collaborative features. The variational collaborative decoder $p_{\theta_r}(R|U, I)$ for user-item interactions R employs a self-supervised learning-based graph neural network for modeling.

2) *Inference Process*: Given the nonlinearity of the generative processes involving hidden variables, which are parameterized as deep neural networks, the exact computation of posterior distributions for these hidden variables becomes intractable. Consequently, we resort to variational inference [29], employing a variational posterior $q(\cdot)$ that is parameterized by deep neural networks to approximate the true posterior. According to conditional independence, the joint posterior of all hidden variables can be factorized into the product of three compact parts:

$$\begin{aligned} q_\phi(Z_{1:M}, M'_{1:K}, Z, U, I|R, X_{1:M}) \\ = \prod q_{\phi_u}(U|R)q_{\phi_i}(I|R)q_{\phi_z}(Z_{1:M}|X_{1:M}) \\ p(M'_{1:K}|Z_{1:M})p(Z|M'_{1:K}, Z_{1:M}), \end{aligned} \quad (6)$$

where $q_{\phi_u}(U|R)$, $q_{\phi_i}(I|R)$ and $q_{\phi_z}(Z_{1:M}|X_{1:M})$ are modeled by inference networks with ϕ_u , ϕ_i and ϕ_z to be trainable parameters.

Following [29], which proves that minimizing the Kullback-Leibler (KL)-divergence between the approximated posterior and the true posterior is equivalent to maximizing the evidence lower bound (ELBO) objective function, we formalize the ELBO as:

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q_\phi} \left[\frac{p_\theta(Z_{1:M}, M'_{1:K}, Z, U, I, R, X_{1:M})}{q_\phi(Z_{1:M}, M'_{1:K}, Z, U, I|R, X_{1:M})} \right], \quad (7)$$

where we resort to variational inference with an approximated posterior distribution as $q_\phi(Z_{1:M}, M'_{1:K}, Z, U, I|R, X_{1:M})$ with variational parameters ϕ .

Furthermore, by substituting the factorized posterior back into the ELBO, we could obtain the new ELBO as follows:

$$\begin{aligned} \mathcal{L}_{ELBO} = \mathbb{E}_{q_{\phi_{ui}}(U, I|R)} [\log p_\theta(R|U, I)] \\ + \mathbb{E}_{q_{\phi_z}(Z|X_{1:M})} [\log p_\theta(X_{1:M}|Z)] \\ - KL(q_{\phi_u}(U|R)||p(U)) - KL(p(M'_{1:K}|Z_{1:M})||p(M'_{1:K})) \\ - KL(q_{\phi_z}(Z_{1:M}|X_{1:M})||p(Z_{1:M})) \\ + \mathbb{E}_{q_{\phi_z}(Z|X_{1:M})} [\log p_\theta(I|Z)]. \end{aligned} \quad (8)$$

Here, the objective function comprises three distinct terms. The first term is the expectation term of the log-likelihood, aimed at compelling the hidden variables to reconstruct the observed input optimally, thereby ensuring the representation learning ability of the model. Specifically, $\mathbb{E}_{q_{\phi_{ui}}(U, I|R)} [\log p_\theta(R|U, I)]$ is the reconstruction loss for user-item ratings, while $\mathbb{E}_{q_{\phi_z}(Z|X_{1:M})} [\log p_\theta(X_{1:M}|Z)]$ is for item multi-modal features. The second term encompasses KL divergence terms and the third term, $\mathbb{E}_{q_{\phi_z}(Z|X_{1:M})} [\log p_\theta(I|Z)]$ forms a constraint for joint multi-modal variables and item collaborative variables.

3) *Kullback-Leibler (KL) Divergence Terms*: The first KL divergence term $KL(q_{\phi_u}(U|R)||p(U))$ regularizes the variational posterior over user latent embeddings to be close to the prior distribution $p(U)$. Specifically, $p(U)$ is set to a unit Gaussian, which constrains excessive informative (noise) from being encoded into user latent embedding $\mathbf{u}$. The second KL term $KL(q_{\phi_z}(Z_{1:M}|X_{1:M})||p(Z_{1:M}))$ prevents the deviation

of the variational posterior $q_{\phi_z}(Z|X)$ from the standard Gaussian prior $p(Z)$ for each modality. For the third KL loss term $KL(p(M'_{1:K}|Z_{1:M})||p(M'_{1:K}))$, the parameters of the conditional distribution $p(M'_{1:K}|Z_{1:M})$ are set to be proportional to the precision (inverse variance) of the input modalities $\mathbf{z}_{1:M}$. In this way, the selection probability of each modality is the inverse of its estimated variance, where the modality with smaller uncertainty would be assigned a greater sampling probability and contribute more to the joint latent variable. Then, the third KL loss term punishes the posterior of $p(M'|Z)$ to be uniformly distribution.

4) *Neural Network Implementation*: We implement two neural networks for user-item interactions and item multi-modal features, which are then coupled in the optimization. For user-item interactions, inspired by BM3 [8], we leverage graph-based contrastive learning for implementation. To reduce the computational complexity and the memory cost, BM3 removes the requirement of graph augmentations with a simple latent embedding dropout technique, where the contrastive latent embeddings $\hat{\mathbf{u}}$ and $\hat{\mathbf{i}}$ of user and item embeddings $\mathbf{u}$ and $\mathbf{i}$ under a dropout ratio p is calculated as:

$$\hat{\mathbf{u}} = \mathbf{u} \cdot \text{Bernoulli}(p), \hat{\mathbf{i}} = \mathbf{i} \cdot \text{Bernoulli}(p), \quad (9)$$

where stop-gradient is conducted upon the contrastive views $\hat{\mathbf{u}}$ and $\hat{\mathbf{i}}$. The original embeddings $\mathbf{u}$ and $\mathbf{i}$ after two multilayer perceptrons (MLPs) form the online representations $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{i}}$ to be optimized. Then, negative cosine similarity between the online embedding and target one of users and items are applied to reconstruct the ratings as:

$$\mathcal{L}_{\text{rec}} = C(\tilde{\mathbf{u}}, \hat{\mathbf{i}}) + C(\tilde{\mathbf{i}}, \hat{\mathbf{u}}). \quad (10)$$

To make predictions, we calculate the similarity between user u and item i using the dot product between their online representations as $\tilde{\mathbf{u}} \cdot \tilde{\mathbf{i}}$.

For item multi-modal features, we employ distinct variational encoders reparameterized by two multiple-layer perceptions (MLP) to embed diverse modality features. Subsequently, K indicators are sampled from Categorical distributions, where the parameters are set to be proportional to the precision (inverse variance) of the input modalities $\mathbf{z}_{1:M}$. The mixture of sampled modalities forms the joint variable, which is then transformed by different variational decoders comprising two MLP layers for the reconstruction of different modality features.

C. Maximum a Posteriori Estimation

Maximum a posteriori estimation can be performed by considering the variational distributions $q_{\phi_{ui}}(U, I|R)$ and $q_{\phi_z}(Z|X_{1:M})$, as well as maximizing the evidence lower bound (ELBO) objective with respect to $\mathbf{u}$, $\mathbf{i}$, and $\mathbf{z}$ by an EM-style algorithm using block coordinate ascent. Since each variable in our paper is defined as a Gaussian distribution, the negative log-likelihood of the Gaussian distribution reduces to an L2-norm loss.

We first optimize the parameters related to collaborative variables $(\mathbf{u}, \mathbf{i})$ while fixing the parameters related to item multi-modal variables $(\mathbf{z}, \mathbf{z}_{1:M}, \mathbf{m}_{1:K})$. The resulting objective for $\mathbf{u}, \mathbf{i}$

thus becomes:

$$\begin{aligned} \mathcal{L}_{ELBO} = & \mathbb{E}_{q_{\phi}(U, I|R)} [\log p_{\theta}(R|U, I)] \\ & - KL(q_{\phi_u}(U|R)||p(U)) - \frac{\lambda_C}{2} \mathbb{E}_{q_{\phi_i}(I|R)} \left\| I - \hat{Z} \right\|_F^2, \end{aligned} \quad (11)$$

where λ_C is a pre-defined parameter to constrain the degree of alignment enforced between item latent embedding and its joint multi-modal embedding. $\hat{Z}$ denotes the mean vector of the joint variable calculated by the multi-modal encoder. Furthermore, to better enforce consistency between the item collaborative embedding and multi-modal embedding, we introduce an additional reconstruction constraint. Specifically, we augment the training objective with an additional term $p_{\theta_{ui}}(R|U, Z)$ that applies the user-item decoder $p_{\theta_{ui}}(\cdot)$ to predict ratings R from the multi-modal embeddings $\mathbf{z}$. By adding the reconstruction loss, the model is incentivized to infer rating probabilities for each item that are consistent between the collaborative and multi-modal representations. This helps align the embedding spaces and strengthen the consistency of rating inferences made from the collaborative or multi-modal item embeddings. The model is thereby encouraged to learn item representations that contain complementary signals from both modalities while remaining compatible with the rating prediction task.

In this paper, we implement $p_{\theta}(R|U, I)$ inspired by BM3. However, a pairwise loss (e.g., BPR [35]) could also be incorporated during implementation. Specifically, during implementation, we sample (user, item) pairs. For negative item samples, we can construct user-items' pairs using other non-interacted items within the same (user, item) batch during training. In this way, we could simplify computations and save time during data processing by avoiding separate sampling of negative samples.

Then, we fix the parameters of variable $\mathbf{u}, \mathbf{i}$ to optimize the item multi-modal VAE objective. We isolate the terms related to $\mathbf{z}, \mathbf{z}_{1:M}, \mathbf{m}_{1:K}$ and the objective thus becomes:

$$\begin{aligned} \mathcal{L}_{ELBO} = & \mathbb{E}_{q_{\phi}(Z|X_{1:M})} [\log p_{\theta}(X_{1:M}|Z)] \\ & - KL(p(M'_{1:K}|Z_{1:M})||p(M'_{1:K})) \\ & - KL(q_{\phi_z}(Z_{1:M}|X_{1:M})||p(Z_{1:M})) \\ & - \frac{\lambda_C}{2} \mathbb{E}_{q_{\phi_z}(Z|X_{1:M})} \left\| Z - \hat{I} \right\|_F^2. \end{aligned} \quad (12)$$

To fully exploit the ability of collaborative information, we first pre-train the collaborative auto-encoder corresponding to $\mathbb{E}_{q_{\phi_{ui}}(U, I|R)} [\log p_{\theta}(R|U, I)]$, and then train our proposed model in an EM-style way.

D. Reparameterization Trick

We leverage the reparameterization trick [29] to reformulate the sampling process in a manner amenable to gradient-based optimization. Specifically, for the Gaussian latent variables $\mathbf{z}_{1:M}$, we calculate their mean μ_z and logarithm of standard deviation σ_z via the corresponding encoder network, which could then be reparameterized by scaling samples drawn from a fixed Gaussian distribution as:

$$\mathbf{z}([\mu_z, \sigma_z], \epsilon) = \mu_z + \epsilon \odot \sigma_z, \quad (13)$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. Through reparameterization, the sampling operation is decoupled from the deep network parameters, enabling direct backpropagation through the sampling process. As shown in [29], this reparameterization exhibits low variance gradients, allowing stochastic optimization via a Monte Carlo gradient estimator with only a single sample per data point.

Furthermore, conventional sampling from a categorical distribution breaks the computational graph and gradients can't flow through. To circumvent this, we employ the Gumbel-Softmax reparameterization trick [36], providing a continuous relaxation enabling gradient flow. This technique is particularly beneficial in variational auto-encoders and other models incorporating categorical distribution sampling, allowing for gradient backpropagation through discrete stochastic nodes. We define the sampling procedure of the pick-up indicator $\mathbf{m}$ for M modalities as follows:

$$\mathbf{m} = \text{onehot} \left(\underset{i}{\operatorname{argmax}} \{g_i + \log \pi_i\} \right), \quad (14)$$

where π represents probabilities for different categories and g_i are samples from Gumbel(0, 1) noise. Here, g_i can be sampled using the transformation as $-\log(-\log(x_i))$, $x_i \sim \text{Uniform}(0, 1)$, which is designed to get non-fixed results. From (14), the non-differentiable part of the whole process is only argmax , where we can use the differentiable softmax function as an approximation. Finally, the generation formula of the M -dimensional sample is:

$$m_i = \frac{\exp\left(\frac{g_i + \log \pi_i}{\tau}\right)}{\sum_j^M \exp\left(\frac{g_j + \log \pi_j}{\tau}\right)} \quad \text{for } i = 1, \dots, M, \quad (15)$$

where τ denotes a pre-defined softmax temperature. Here, with a smaller τ , softmax approximates tighter to argmax . The training procedure of VMoSE is summarized in Algorithm 1.

E. Extension to Student-T Distribution

We further extend the Gaussian VMoSE to Student-T distribution, which generalizes the normal distribution with heavier tails and could better exhibit the flexibility of our model. We exclusively represent the latent variables of item multi-modal features $\mathbf{z}_{1:M}$ using Student-T distributions as we emphasize on enhancing the modeling capacity of item multi-modal features for the noise-robust recommendation. Specifically, we introduce a Gamma prior distribution that is contingent on the item content variable $\mathbf{x}$, denoted as $\Gamma(\gamma|a(\mathbf{x}), b(\mathbf{x}))$, where $a(\mathbf{x})$ and $b(\mathbf{x})$ represent the shape and rate parameters, respectively. Through the analytical integration of γ , we derive the conditional distribution of the latent content variable $\mathbf{z}$ given the observation $\mathbf{x}$ as follows:

$$\begin{aligned} p_{\theta}(\mathbf{z}|\mathbf{x}) &= \int_0^{\infty} \mathcal{N}(\mathbf{z}|\mu(\mathbf{x}), \gamma^{-1}) \Gamma(\gamma|a(\mathbf{x}), b(\mathbf{x})) d\gamma \\ &= \frac{\Gamma\left(\frac{\nu(\mathbf{x})+1}{2}\right)}{\Gamma\left(\frac{\nu(\mathbf{x})}{2}\right)} \left(\frac{\lambda(\mathbf{x})}{\pi\nu(\mathbf{x})}\right)^{\frac{1}{2}} \left[1 + \frac{\lambda(\mathbf{x})(\mathbf{z} - \mu(\mathbf{x}))^2}{\nu_{\theta}(\mathbf{x})}\right]^{-\frac{\nu(\mathbf{x})+1}{2}} \\ &= \text{St}(\mathbf{z}|\mu_{\theta}(\mathbf{x}), \lambda_{\theta}(\mathbf{x}), \nu_{\theta}(\mathbf{x})), \end{aligned} \quad (16)$$

where $\mu_{\theta}(\mathbf{x})$ denotes the mean vector, $\lambda_{\theta}(\mathbf{x}) = a(\mathbf{x})/b(\mathbf{x})$ represents the precision (inverse of scale) of the Student-T distribution, and $\nu_{\theta}(\mathbf{x}) = 2a(\mathbf{x})$ is the degree of freedom. For $\nu_{\theta}(\mathbf{x})$ approaches $+\infty$, the Student-T distribution tends towards a Gaussian distribution of the same mean vector and precision matrix. We utilize $a(\mathbf{x})/b(\mathbf{x})$ to be precision and sample modalities from the Categorical distribution of the normalized precision.

To calculate the objective functions in our model, it is necessary to evaluate the term $KL(q_{\phi_z}(Z_{1:M}|X_{1:M})||p(Z_{1:M}))$. However, since there is no analytical solution for the KL divergence between Student-T distributions, we draw inspiration from the work in [37]. It introduces an approximation method that leverages a Taylor expansion to estimate the cross entropy, yielding an asymptotic form of the KL divergence. This approximation converges towards a closed-form solution, analogous to the case involving a normal distribution, particularly when the degrees of freedom approach $+\infty$. We employ this method as follows:

$$\begin{aligned} &KL(\text{St}(\mu_1, \lambda_1^{-1}, \nu_1) || \text{St}(\mu_2, \lambda_2^{-1}, \nu_2)) \\ &= \frac{1}{2} \left\{ C_2 (\mu_2 - \mu_1)^{\top} \lambda_2^{-1} (\mu_2 - \mu_1) + C_1 C_2 \text{rank}(\lambda_2 \lambda_1^{-1}) \right. \\ &\quad \left. - \log |\lambda_2| |\lambda_2|^{-1} - d_1 \right\}, \end{aligned} \quad (17)$$

where,

$$\begin{aligned} C_1 &= \nu_1 / (\nu_1 - 2), C_2 = (\nu_2 + d) / \nu_2, \\ d_1 &= (\nu_1 + d) \{ \psi((\nu_1 + d)/2) - \psi(\nu_1/2) \}. \end{aligned} \quad (18)$$

Here, $\text{rank}()$ calculates the rank of the matrix and $\psi()$ is the digamma function.

Furthermore, the reparameterization trick needs to be applied to the Student-T distribution. This ensures that the sampling of the expectation process occurs outside the model, rendering it amenable to gradient-based optimization. The procedure is as follows:

$$\mathbf{z}([\mu_z, \lambda_z, \nu_z], \epsilon') = \mu_z + \epsilon' \odot \lambda_z, \epsilon' \sim \text{St}(\mathbf{0}, \mathbf{I}_d, \nu_z). \quad (19)$$

However, the resulting reparameterized distribution is not a differentiable transformation. An alternative reformulation of the Student-T distribution was proposed by [38], defined as: let $n \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and $g \sim \Gamma(\frac{\nu}{2}, 0.5)$, then $\epsilon' = n/\sqrt{\frac{g}{\nu}}$ follows a Student-T distribution with $\epsilon' \sim \text{St}(\mathbf{0}, \mathbf{I}_d, \nu)$. Leveraging this formulation, [39] derives an implicit differentiation of the cumulative distribution function of the Gamma distribution. With this reformulation of the Student-T distribution in terms of Gaussian and Gamma random variables, the distribution becomes fully differentiable with respect to all three parameters μ, λ , and ν . This enables direct backpropagation through samples from the Student-T distribution, allowing end-to-end training of neural networks involving this distribution. The reformulation in terms of more elementary distributions facilitates differentiation, unlocking broader use of the flexible Student-T for robust modeling of non-Gaussian posteriors in variational inference frameworks.

TABLE II
STATISTICS OF DATASETS AFTER PREPROCESSING

Dataset	#Users	#Items	#User-Item	Sparsity
Movies	14,998	4,266	80,114	0.125%
Electronics	41,138	14,845	324,746	0.053%
Sports	35,598	18,357	296,337	0.045%

F. Prediction

Given the trained VMoSE model, the weights of the inference networks and generative networks for both the item implicit feedback and multi-modal content are learned. For a new prediction, the expected rating $\hat{r}$ can then be computed as:

$$\mathbb{E}[\hat{r}|D] = p_{\theta_v}(\hat{r}|\mu_i, \mu_u) + p_{\theta_r}(\hat{r}|\mu_i, \mu_z), \quad (20)$$

where μ_i , μ_u and μ_z denotes the mean vector of the latent item variable, user variable, and joint multi-modal variable through the collaborative encoder and item multi-modal encoder. D is the training set. The term $p_{\theta_r}(\mathbf{r}|\mu_i, \mu_z)$ applies the collaborative decoder $p_{\theta_r}(\cdot)$ to predict ratings from the inferred mean multi-modal latent variable μ_z . This reconstruction from the joint multi-modal embedding further enforces the role of the multi-modal features in the rating prediction.

G. Computational Complexity

The time complexity of VMoSE primarily comprises three components: 1) Graph convolution: following LightGCN [40] and BM3 [8], we adopt linear propagation of the normalized adjacency matrix, which constitutes a substantial portion of the overall computation. Given L as the number of LightGCN layers, $|\mathcal{V}|$, $|\mathcal{U}|$, $|\mathcal{I}|$ and $|\mathcal{E}|$ as the number of nodes, users, items and edges in the user-item graph, the analytical complexity of graph convolution is $\mathcal{O}(L|\mathcal{E}|d)$. 2) Feature transform: we utilize additional operations on multi-modal feature projection, with a complexity of $\mathcal{O}(\sum_{m \in \mathcal{M}} |\mathcal{I}|d_m d)$, where d_m denotes the original feature dimensions of the m th modality. Specifically, the sampling process during the multi-modal fusion takes $\mathcal{O}(|\mathcal{I}|d)$, which is negligible compared to the feature transformation operation. 3) Loss function: the cost of contrastive loss component is $\mathcal{O}(|\mathcal{M}||\mathcal{E}|d)$. Additionally, the cost of reconstruction loss, KL divergence loss, and MSE loss is $\mathcal{O}(\sum_{m \in \mathcal{M}} |\mathcal{I}|d_m d + 3|\mathcal{I}|d)$. Therefore, the total computational cost of VMoSE is $\mathcal{O}(L|\mathcal{E}|d + 2\sum_{m \in \mathcal{M}} |\mathcal{I}|d_m d + |\mathcal{M}||\mathcal{E}|d + 3|\mathcal{I}|d)$, which is in the same magnitude as BM3.

However, compared with MMGCN, MVGAE and DualGNN, which employ different GCNs for distinct multi-modal features and a two-layer MLP for multi-modal feature projection, the computational cost for graph convolution and feature transform are $\mathcal{O}(|\mathcal{M}||\mathcal{L}||\mathcal{E}|d)$ and $\mathcal{O}(\sum_{m \in \mathcal{M}} |\mathcal{I}|(d_m + d)d_h)$, respectively. Here, d_h denotes the dimension of the hidden layer in a two-layer MLP. In comparison to LATTICE and MICRO, which construct an item-item graph from multi-modal features, incurring costs of $\mathcal{O}(|\mathcal{I}|^2 d_m)$ to build the similarity matrix between items, $\mathcal{O}(|\mathcal{I}|^3)$ to normalize the matrix, and $\mathcal{O}(k|\mathcal{I}|\log(|\mathcal{I}|))$ to retrieve top- k most similar items for

Algorithm 1: Training VMoSE With SGD.

Input: User-Item interaction matrix $\mathbf{R}$; Item multi-modal content matrix $\mathbf{X}_{1:M}$

Randomly initialize θ , ϕ .

while not converged do

// Update item collaborative information.

forall $(u, i) \in \mathbf{R}$ **do**

Randomly sample a batch of user-item pairs.

Infer mean and variance of the user and item latent variables $\mathbf{u}, \mathbf{i}$.

Sample $\mathbf{u}, \mathbf{i}$ via reparametrization trick.

Get mean of multi-modal joint latent embedding $\hat{\mathbf{z}}$.

Compute the item loss via Eq. (11).

end

Compute the gradient of the item loss.

Update θ_r, ϕ_{ui} by taking stochastic gradient steps.

// Update multi-modal Content information.

forall $v \in \mathcal{V}$ **do**

Randomly sample a batch of items with $\mathbf{X}_{1:M}$.

Infer mean and variance of the modality-specific latent variables $\mathbf{z}_{1:M}$.

Sample $\mathbf{m}_{1:K}$ via Gumbel-softmax reparametrization trick.

Sample $\mathbf{z}_{1:M}$ via reparametrization trick.

Get multi-modal joint latent embedding $\mathbf{z}$ via Eq. (1).

Get mean of item latent variable $\hat{\mathbf{i}}$.

Compute the content loss via Eq. (12).

end

Compute the gradient of the content loss.

Update θ_x, ϕ_z by taking stochastic gradient steps.

end

return θ, ϕ

each item. For SLMRec and MMSSL, which employ self-augmentation and multi-task strategy for training, the additional cost for data augmentation is $\mathcal{O}(|\mathcal{E}||\mathcal{M}||\mathcal{I}|d)$. The InfoNCE-based contrastive learning requires $\mathcal{O}(|\mathcal{M}||\mathcal{E}|d)$ and $\mathcal{O}(|\mathcal{M}||\mathcal{E}||\mathcal{U}|d)$ cost for numerator and denominator calculation, respectively. In summary, the computational cost of VMoSE is comparable to BM3, making it more efficient than other vanilla graph-based methods and graph-based contrastive learning methods. Compared to traditional graph-based methods (MMGCN, MVGAE, and DualGNN), we use the simplified LightGCN version and remove the feature transformation operation on the graph. Compared with the graph-based contrastive learning methods (SLMRec, MMSSL, LATTICE, and MICRO), we use a stop-gradient strategy for augmentation, eliminating the need for additional InfoNCE loss calculation, and does not require additional construction of item-item graph as additional input.

IV. EXPERIMENTS

To evaluate the efficacy of our proposed multi-modal recommendation model, we conduct a series of experiments utilizing three datasets. The evaluation process involves comparing the performance of various multi-modal recommendation techniques. Furthermore, we carry out an ablation study to confirm the effectiveness of stochastic sampling, in which we compare different numbers of sampling to validate the proposed multi-modal fusion strategy. We also investigate the effectiveness of multi-modal approaches compared to using only single-modal information. Finally, we assess the robustness of our model by evaluating its ability to handle modality missing and noise. Meanwhile, we visualize the distribution of the modality sampling and verify whether our model can reasonably model the uncertainty levels of different modalities and then utilize the uncertainty to perform sampling-based fusion, achieving better robustness.

A. Datasets

Throughout this section, we make use of the following datasets: Movies and TV (abbreviated to Movies), Electronics, and Sports and Outdoors (abbreviated to Sports). They originate from subareas of Amazon review.¹ Statistics of the datasets are summarized in Table II, where the data sparsity is measured as the number of interactions divided by the product of the number of users and the number of items. We use a 3-core setting for Movies and 5-core settings for Electronics and Sports respectively to ensure data quality and volume. The pre-processed datasets include both visual and textual modalities. For Movies and Electronics, pre-trained ResNet50 [10] and Sentence2Vec [41] are used to extract corresponding visual and textual features and obtain 2,048 and 100-dimensional embeddings. For Sports, following the approach outlined in [8], we employ the Caffe reference model² [42], [43] (referred to as Deep CNN) pre-trained on 1.2 million ImageNet (ILSVRC2010) images. This model generates 4,096-dimensional visual features. Textual embeddings are obtained by concatenating the title, descriptions, categories, and brand of each item, and Sentence-transformers [44] are employed to produce 384-dimensional sentence embeddings for the textual modality.

B. Evaluation Protocols

To enable a fair comparison, we adopt the same evaluation protocol utilized in prior works [8], [9], randomly splitting the interaction history of each user into training, validation, and testing sets with a ratio of 8:1:1. Moreover, we employ Recall@ p and NDCG@ p to evaluate the top- p recommendation performance. Specifically, we utilize the all-ranking protocol as opposed to the negative-sampling protocol for computing the evaluation metrics, as this allows for a more comprehensive assessment of recommendation accuracy. In the recommendation phase, all items uninteracted by the given user are treated as candidate items.

We report the experimental results for $p = 10$ and $p = 20$, abbreviating the metrics Recall@10, Recall@20, NDCG@10, and NDCG@20 as R@10, R@20, N@10, and N@20, respectively.

Specifically, Recall@ p calculates the proportion of relevant items (hits) within the top- p recommendations out of all relevant items for a user u , defined as:

$$\text{Recall}_u@p = \frac{|r_u \cap y_u|}{|y_u|} \quad (21)$$

$$\text{Recall}@p = \frac{\sum_{u \in \mathcal{U}^{te}} \text{Recall}_u@p}{|\mathcal{U}^{te}|}, \quad (22)$$

where r_u represents the set of top p recommended items for user u , y_u denotes the set of interacted items in the test set, and $|\cdot|$ denotes the number of the set. $\mathcal{U}^{te}$ represents the set of users in the test set, $|\mathcal{U}^{te}|$ represents the number of users in the test set. Hence, Recall@ p calculates the number of hit items in the top- p recommendation list among Ground Truth of user u and Recall@ p averages Recall@ p to calculate the hit ratio of all users in the test set.

Normalized Discounted Cumulative Gain (NDCG) accounts for the position of hits by assigning higher scores to hits at higher ranks in the top- p recommendations, calculated as:

$$\text{NDCG}_u@p = \left(\sum_{i=1}^p \frac{2^{r_i} - 1}{\log_2(i + 1)} \right) / S_u \quad (23)$$

$$\text{NDCG}@p = \frac{\sum_{u \in \mathcal{U}^{te}} \text{NDCG}_u@p}{|\mathcal{U}^{te}|}, \quad (24)$$

where r_i is the relevance of the item at position i (binary 0 or 1), and S_u is a normalization constant equal to $\frac{|y_u|}{\log_2}$, which guarantees the NDCG@ p ratio always to be in the range [0, 1]. It means that when the item in the top- p list hits the Ground Truth y_u , the discounted cumulative gain (DCG) score would increase according to its ranking position.

C. Baselines

- **VBPR [45]:** It integrates visual features and latent identifier factors (non-visual) within the framework of matrix factorization (MF) for collaborative filtering and employs a Bayesian personalized ranking loss function.
- **MMGCN [22]:** It incorporates multi-modal features and implements distinct graph convolutional networks for each modality. The embedding representations learned from the various modalities are then averaged to obtain the final unified representations for the user and item nodes.
- **CVAE [33]:** This is a variational model that accounts for noise in the feature representations of items. It learns latent content variables through variational autoencoders, while also incorporating collaborative filtering variables that leverage the information from user-item interactions.
- **MVGAE [1]:** It is a variational model that implements separate graph convolutional network (GCN) encoders to encode each modality and derive modality-specific latent variables. It then fuses these variables using a product-of-experts principle to obtain a joint latent representation for generating recommendations.

¹<http://jmcauley.ucsd.edu/data/amazon/>

²bvlc reference caffe net from caffe.berkeleyvision.org

TABLE III
OVERALL COMPARISONS WITH DIFFERENT MULTI-MODAL RECOMMENDATION MODELS

	Movies				Electronics				Sports			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
VBPR [45]	0.1682	0.1009	0.2335	0.1177	0.0307	0.0186	0.0471	0.0234	0.0278	0.0160	0.0425	0.0198
MMGCN [22]	0.1692	0.0975	0.2383	0.1132	0.0318	0.0200	0.0530	0.0260	0.0370	0.0193	0.0605	0.0254
CVAE [33]	0.1713	0.0771	0.2525	0.0984	0.0406	0.0222	0.0663	0.0295	0.0335	0.0177	0.0507	0.0216
MVGAE [1]	0.1732	0.1013	0.2534	0.1194	0.0428	0.0266	0.0673	0.0337	0.0436	0.0242	0.0710	0.0314
DualGNN [23]	0.2021	0.1292	0.2765	0.1485	0.0519	<u>0.0330</u>	0.0795	<u>0.0409</u>	0.0568	0.0310	0.0859	0.0385
SLMRec [24]	0.1693	0.0913	0.2529	0.1128	0.0446	0.0278	0.0680	0.0345	0.0470	0.0255	0.0829	0.0376
LATTICE [28]	0.1336	0.0711	0.2301	0.0966	0.0428	0.0244	0.0704	0.0320	0.0618	0.0337	0.0915	0.0424
BM3 [8]	<u>0.2311</u>	<u>0.1379</u>	<u>0.3092</u>	<u>0.1581</u>	<u>0.0553</u>	0.0321	0.0810	0.0388	<u>0.0656</u>	<u>0.0355</u>	0.0980	0.0438
MMSSL [9]	0.1992	0.1210	0.2793	0.1425	0.0476	0.0297	0.0738	0.0372	0.0505	0.0278	0.0998	0.0470
MICRO [46]	0.1529	0.0854	0.2521	0.1091	0.0474	0.0274	0.0752	0.0349	0.0556	0.0300	<u>0.1005</u>	0.0467
Ours_Gau	0.2374	0.1424	0.3202	0.1638	0.0599	0.0344	0.0900	0.0422	0.0666	0.0369	0.1012	0.0457
Ours_T	0.2378	0.1426	0.3211	0.1641	0.0604	0.0343	0.0900	0.0420	0.0671	0.0371	0.1010	0.0457

- *DualGNN* [23]: It is a micro-video recommendation model, that builds user co-occurrence graphs and capitalizes on the inter-user correlations to collaboratively extract specific fusion patterns tailored to each user.
- *SLMRec* [24]: This is a generic framework using data augmentation upon multi-modal contents and contrastive learning which differentiates the views of an item from the others’.
- *LATTICE* [28]: It learns latent item-item semantic structures from multi-modal features through modality-aware graph learning and convolutions to enrich item representations for improved recommendations.
- *BM3* [8]: It jointly optimizes three objectives to learn user and item representations by reconstructing the user-item graph and aligning multi-modal features, without needing negative samples or complex graph augmentations for contrastive view generation.
- *MMSSL* [9]: This is a self-supervised learning method for recommendation that uses modality-aware interactive structure learning via adversarial perturbations for data augmentation and cross-modal contrastive learning to jointly preserve inter-modal semantics and user preference diversity without labels.
- *MICRO* [46]: It is an extension of *LATTICE*, wherein latent item relations are derived through the utilization of the item-item graph. Subsequently, fine-grained multi-modal fusion is executed based on the structures identified through the item-item graph mining process.

D. Implementation Details

For a fair comparison with other works, we fix the embedding size to 64 for all models. For our model, we use Adam as the optimizer and search for the optimal learning rate in [0.01, 0.05, 0.001, 0.005, 0.0001, 0.0005]. Specifically, we first pre-train the collaborative model (using the variational version of BM3 without features), with 2 layers of GCN and dropout searched in [0.2, 0.3]. We use 1e-3 as the learning rate and weight decay for pre-training. Then, we train our proposed model, where we search for the optimal parameters for the multi-modal autoencoder, and set a small learning rate as 1e-5 for the collaborative model, with the weight of the similarity constraint

searched in [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]. After the search, we set the learning rate for the multi-modal model as 1e-3 for Electronics and Sports datasets and 1e-5 for Movies. For the similarity constraint, we set 2, 4, and 5 for Movies, Electronics, and Sports, respectively.

E. Overall Comparisons

The overall performance of our proposed VMoSE and the state-of-the-art baselines is summarized in Table III. Results highlighted in bold indicate the best performance across all methods, while the results underlined signify the top performance amongst the baseline models, excluding our proposed approach. From the table, we have the following observations:

- Graph-based models (MMGCN, VMGAE, and DualGNN) outperform MF-based methods (VBPR and CVAE) by modeling high-order connectivity between users and items via direct graph links as well as multiple hops along the graph. However, CVAE uses a variational-based training strategy to filter out irrelevant noises, achieving better performance than MMGCN which only uses GCN for feature and interaction modeling. VMGAE improves over MMGCN by using a PoE fusion approach, assigning more weights to modalities containing more recommendation signals, thus achieving better performance than MMGCN and CVAE. DualGNN further incorporates an extra user-user correlation graph derived from the user-item bipartite graph. Notably, superior outcomes, particularly in the Electronics dataset, are attained through the fusion of user representations obtained from neighboring nodes in the correlation graph.
- Furthermore, contrastive learning-based methods (SLMRec, LATTICE, BM3, MMSSL, and MICRO) can achieve even better performance over basic GNN-based methods by training representations via maximizing agreement between differently augmented views of the same data example via a contrastive loss. SLMRec uses data augmentation upon item multi-modal features and contrastive loss between different views of items. LATTICE further models graph relations between multi-modal item-item interactions and achieves better performance over SLMRec on sparser Electronics and Sports datasets.

TABLE IV
 COMPARISONS WITH DIFFERENT SAMPLING NUMBERS

	Movies				Electronics				Sports			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
Ours_Gau												
K=0	0.2358	0.1416	0.3184	0.1629	<u>0.0602</u>	0.0342	0.0894	0.0418	<u>0.0669</u>	0.0370	0.1007	0.0456
K=1	0.2379	0.1423	0.3190	0.1632	0.0599	0.0344	0.0900	0.0422	0.0668	0.0371	0.1007	0.0457
K=2	0.2374	0.1424	<u>0.3202</u>	<u>0.1638</u>	0.0598	0.0342	0.0896	0.0421	0.0666	0.0369	0.1012	0.0457
K=3	0.2382	0.1428	0.3189	0.1637	0.0601	0.0343	0.0897	0.0420	0.0668	0.0371	0.1007	0.0457
Ours_T												
K=0	0.2358	0.1423	0.3198	<u>0.1641</u>	0.0603	0.0342	0.0896	0.0420	0.0667	0.0368	0.0999	0.0452
K=1	0.2364	0.1423	0.3195	0.1638	0.0604	0.0344	0.0898	<u>0.0421</u>	0.0658	0.0366	0.0995	0.0452
K=2	0.2378	<u>0.1426</u>	0.3211	0.1641	0.0604	0.0343	0.0900	0.0420	0.0671	0.0371	0.1010	0.0457
K=3	0.2374	0.1425	0.3200	0.1639	0.0601	0.0343	0.0898	0.0420	0.0665	0.0369	0.1002	0.0454

Building upon LATTICE, BM3, and MMSSL improve over LATTICE by designing lighter feature dropout as contrastive views and finer-grained adversarial perturbations for augmentation respectively. MICRO, as an extension of LATTICE, achieves even better results on the Sports dataset by forcing the fused multi-modal representations to adaptively capture item relationships shared between multiple modalities in a self-supervised manner.

- Our proposed VMoSE model learns ID embeddings with contrastive learning using lightweight feature dropout for user-item interactions. It also designs a multi-modal fusion model using modality sampling to learn multi-modal embeddings, assigning less weights to noisier modalities. SmoothL1Loss is used to constrain the similarity between ID and multi-modal embeddings. Thus the fused multi-modal features contain more recommendation-relevant signals, achieving better performance. Building upon this foundation, we implement latent variable modeling based on the Gaussian and Student-T distributions, denoted as Ours_Gau and Ours_T, respectively. From the results, we could see that our model could achieve better performance among other baselines. Furthermore, Student-T based implementation achieves improved performance on certain datasets, which demonstrates greater robustness to noise in the tail ends of long-tail distributions compared to Gaussian modeling. In particular, VMoSE improves Recall@10 over the strongest baseline by 2.90%, 9.22%, and 2.29% on the three datasets.

In summary, the results validate the benefits of VMoSE's probabilistic multi-modal fusion approach. By modeling uncertainty and constraining ID similarity, it filters noise and focuses on informative signals. The performance gains over state-of-the-art GNN and contrastive learning methods showcase the advantages of our proposed fusion strategy. The experiments demonstrate VMoSE's ability to effectively integrate multi-modal data for improved recommendation quality.

F. Effectiveness of Stochastic Sampling

1) *Comparisons With Different Sampling Numbers:* To explore the efficacy of modality sampling, we compare the

performance without modality sampling ($K=0$) and with different numbers of sampled modalities ($K=1, 2, 3$). Specifically, for $K=0$, we replace the modality sampling strategy with a simple averaging of the two modality features. For Gaussian and Student-T embeddings, we underline the respective optimal values, and in addition, we use bold to indicate the overall optimal value between the two. As shown in Table IV, we can see that the fusion strategy with modality sampling outperforms the scenario where modality fusion is not utilized. In particular, for Movies and Electronics datasets that have higher feature noise, direct averaging performs even worse than sampling with just one modality, demonstrating the efficacy of the proposed modality fusion. On the other hand, for all three datasets, using two (Movies and Sports) or one (Electronics) modality sampling generally works better than using too many sampled modalities ($K=3$). The possible reason for the reduced performance is that excessive sampling from the two modalities introduces greater noise as well as redundancy, which impedes the model's ability to learn effective representations.

Furthermore, we could see that optimal K values may vary across different datasets and different distribution embeddings (Gaussian and Student-T). For experimental consistency, we selected $K=2$ for default, and $K=1$ for Electronics dataset with Gaussian embeddings based on the parameter search process, a value that demonstrated good performance across diverse metrics and achieved best results on Recall@20.

2) *Comparisons With Single Modality:* To further validate the efficacy of our modality fusion strategy, we conduct comparisons with method that do not utilize any modality features (referred to as "without"), methods employing only the visual modality or the textual modality, and our proposed multi-modal method. Specifically, the "without" method exclusively employs our pre-trained interaction model, while the single modality methods omit the modality sampling and fusion modules with one modality. For Gaussian and Student-T embeddings, we underline the respective optimal values, and in addition, we use bold to indicate the overall optimal value between the two. As shown in Table V, we can see that the multi-modal fusion strategy overall outperforms both the without modality and single modality methods, demonstrating the efficacy of our proposed modality sampling approach. Moreover, visual information is typically more intuitive and engaging to users than textual

TABLE V
RESULTS BETWEEN SINGLE-MODAL AND MULTI-MODAL MODELS

	Movies				Electronics				Sports			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
Ours_Gau												
without	0.2315	0.1379	0.3167	0.1599	0.0585	0.0332	0.0885	0.0410	0.0629	0.0341	0.0942	0.0421
Visual	0.2358	0.1411	0.3183	0.1625	<u>0.0601</u>	0.0342	0.0899	0.0419	<u>0.0671</u>	<u>0.0371</u>	0.0998	0.0454
Textual	0.2360	0.1417	0.3179	0.1629	0.0600	0.0342	0.0897	0.0419	0.0670	0.0368	0.1000	0.0452
Multi-modal	<u>0.2374</u>	<u>0.1424</u>	<u>0.3202</u>	<u>0.1638</u>	0.0599	0.0344	0.0900	0.0422	0.0666	0.0369	0.1012	0.0457
Ours_T												
without	0.2315	0.1379	0.3167	0.1599	0.0585	0.0332	0.0885	0.0410	0.0629	0.0341	0.0942	0.0421
Visual	0.2375	0.1430	0.3198	0.1643	0.0602	0.0342	0.0897	0.0419	0.0666	0.0369	0.1003	0.0455
Textual	0.2364	0.1415	0.3193	0.1630	0.0600	0.0341	0.0896	0.0418	0.0666	0.0369	0.1002	0.0454
Multi-modal	0.2378	0.1426	0.3211	0.1641	0.0604	<u>0.0343</u>	0.0900	<u>0.0420</u>	0.0671	0.0371	<u>0.1010</u>	0.0457

TABLE VI
RESULTS ON MODALITY MISSING PROBLEM DURING TRAINING

	Movies				Electronics				Sports			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
Ours_Gau												
V-missing	0.2375	0.1423	0.3192	0.1634	0.0603	0.0344	0.0897	0.0420	0.0667	0.0369	0.1005	0.0455
T-missing	0.2377	0.1426	0.3193	0.1637	0.0600	0.0344	0.0899	0.0423	0.0668	0.0369	0.1009	0.0456
No missing	0.2374	0.1424	0.3202	0.1638	0.0599	0.0344	0.0900	0.0422	0.0666	0.0369	0.1012	0.0457
Ours_T												
V-missing	0.2366	0.1424	0.3187	0.1637	0.0602	0.0343	0.0895	0.0420	0.0665	0.0368	0.1001	0.0453
T-missing	0.2380	0.1427	0.3186	0.1635	0.0601	0.0343	0.0897	0.0420	0.0663	0.0367	0.0998	0.0452
No missing	0.2378	0.1426	0.3211	0.1641	0.0604	0.0343	0.0900	0.0420	0.0671	0.0371	0.1010	0.0457

V and T represent visual and textual, respectively.

descriptions, especially with Student-T embeddings, providing more cues related to clicks. Thus using only the visual modality performs better than using just the textual modality.

G. Robustness to Modality Missing

Modality missing is inevitable sometimes in real-world scenarios due to data collection issues or user-side omissions. To simulate missing modalities during training, we mask different ratios of the visual (or textual) features. For items with both modalities, we apply multi-modal sampling and fusion. For items with only a single modality, we use only the remaining modality. Except for the noted manipulations, all other aspects of the experimental design were equated between groups. Results of 50% modality missing are listed in Table VI. From the results, we can see that missing modalities do not significantly hurt performance for both Gaussian and Student-T embeddings. On the Movies and Sports datasets, using one full modality plus 50% of the other modality substantially outperforms using only a single modality shown in Table V. On Electronics, it performs on par with a single modality. These results highlight the robustness of our proposed model in handling missing modalities. Through dynamically fusing available signals, the model demonstrates an ability to marginalize absent modalities and focus inference on remaining informative features. Even when provided with only partial modalities as input, the model retains efficacy in generating high-quality recommendations. Furthermore, we conduct

experiments on different missing ratios in $\{0, 0.2, 0.4, 0.6, 0.8\}$ on Gaussian implementation to better demonstrate the performance of our method. The results of Recall@20 are shown in Fig. 2. The results indicate that, across all three datasets, the absence of the visual modality results in a more significant performance degradation compared to the absence of the textual modality. This phenomenon may be attributed to the model's reliance on visual modalities in recommenders.

H. Robustness to Modality Noise

To verify the robustness of our model to noisy modalities, we add different levels of Gaussian noise to the visual and textual features and evaluate the performance. As shown in Fig. 3, injecting noise does lead to minor performance drops. However, the impact is limited thanks to our multi-modal sampling and fusion approach. Specifically, our method models the uncertainty of each modality and assigns lower sampling probabilities to noisier features. This reduces their influence on the fused multi-modal representation. Thus, even with corrupted visual or textual data, our model can marginalize the noisy signals and rely more on clean modalities. Furthermore, we find that injecting a moderate amount of noise can occasionally improve performance. The added randomness acts as a regularization that prevents overfitting during the variational auto-encoder training. Our model is able to filter out the obvious noise through the fusion process. In summary, by dynamically adjusting sampling

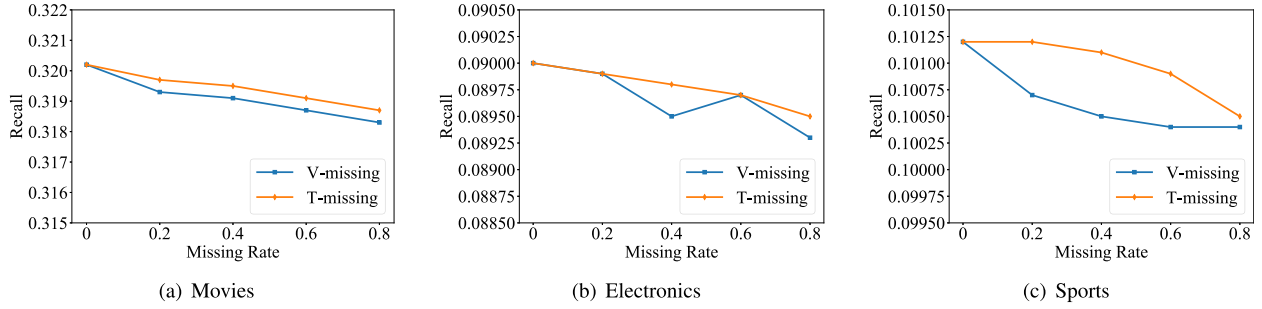


Fig. 2. Effectiveness of modality missing towards different missing rates.

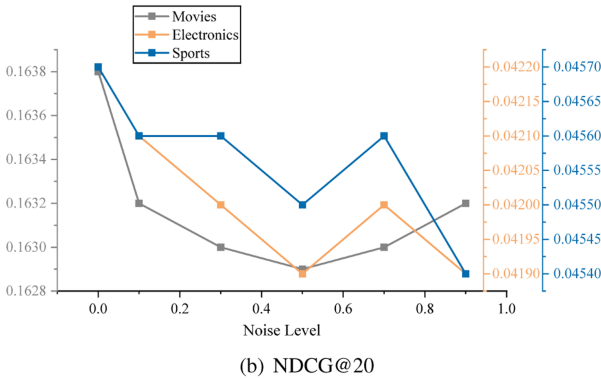
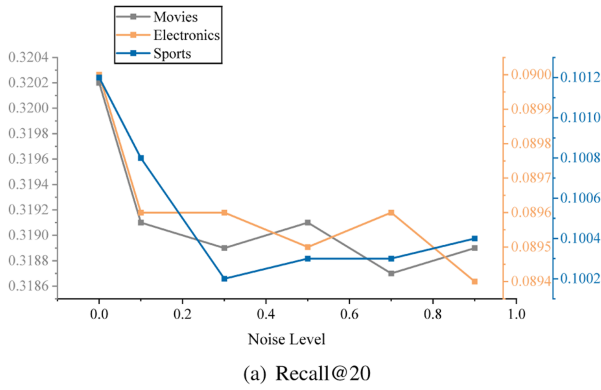


Fig. 3. Impact of noise level towards the results of Recall@20 and NDCG@20 on three datasets.

probabilities based on uncertainty, our method demonstrates strong robustness to corrupted modalities. Even with noisy inputs, our approach can focus on reliable modalities and generate robust recommendations.

I. Effectiveness of Uncertainty Estimation

To validate the accuracy of our uncertainty modeling for different modalities and the effectiveness of sampling based on uncertainty, we add Gaussian noise with magnitudes of 0.1 and 0.9 to the visual features. As depicted in Fig. 4, when a small amount of visual noise ($\epsilon=0.1$) is introduced, the sampling weight assigned to the visual modality outweighs that of text. This outcome aligns with our expectations, given that visual signals offer more informative cues for movie recommendations. However,

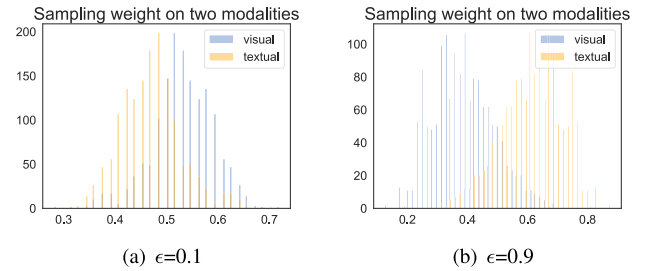


Fig. 4. Effect of noise level towards the weights of visual and textual modalities on Movies datasets.

with the introduction of substantial noise ($\epsilon=0.9$) to corrupt the visual features, the modeled uncertainty associated with the visual modality notably increases, leading to reduced precision. Correspondingly, the sampling probability for the noisy visual features decreases as well. This demonstrates the capacity of our uncertainty-aware sampling method to automatically adapt and diminish the sampling weight for modalities when their reliability is compromised. In summary, our approach accurately models the uncertainty of each modality and dynamically adjusts the sampling probabilities based on the noise levels. When visual information is corrupted with extreme noise, our model automatically falls back to using text features more heavily. This achieves good robustness against noisy or missing modalities by guiding the fusion to focus on more reliable signals.

V. CONCLUSIONS AND FUTURE WORK

In this work, we have proposed a novel variational mixture of stochastic experts (VMoSE) model for a robust multi-modal recommendation by adaptive sampling and fusing of noisy multi-modal signals based on uncertainty estimates. This provides robustness against irrelevant noise and missing modalities. Moreover, we propose a principled and robust solution to challenges like modality noise, missing data, and non-Gaussianity in multi-modal recommendations. By combining the strengths of mixture-of-experts and product-of-experts methods, the stochastic sampling approach provides robustness while retaining expressive latent distributions. Experiments show that the proposed VMoSE approach outperforms existing models for multi-modal recommendation. The added randomness during training further improves robustness to missing modalities at test time.

Future work includes incorporating recent advances in normalizing flows or autoregressive models to increase representation power, integrating the multi-modal fusion approach into a sequential recommender system, and exploring additional applications of uncertainty-based multi-modal fusion.

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